Sprout 1.2 LVSS Skid – Manual Functional Test Procedure (Section 1.1.5)

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Purpose and scope

This procedure performs every Section 1.1.5 functional test by hand, using vendor software, a calibrator and a meter instead of the Python test tool. Use it in three cases:

- **Verification:** spot-check a sample of the tool's results to prove the tool measures correctly (recommended on the first skid and after any tool change).
- **Backup:** run the full test by hand if the laptop or tool is unavailable on test day.
- **Dispute:** re-test any single item the Owner or CxA questions.

Step numbers match the test plan (1.1.5-1.x MOXA IO, 1.1.5-2.x Network Switch, 1.1.5-3.x SPD) and the tool's test IDs, so manual and automated results can be compared line for line. Pass limits come from the Settings tab of the skid config workbook; the values shown here are the current placeholders and must match the approved spec.

Prerequisites

Testing is performed by qualified manufacturer personnel with an Owner or CxA witness where required. All physical SPD work and continuity checks are done de-energized under LOTO per company procedure.

Equipment and software

Item	Used for
Laptop, static IP on the skid subnet, Ethernet cable	All network and IO steps
MOXA ioSearch and ioAdmin	Find modules, view live channel values, view and export channel configuration
Modbus master software (Modbus Poll, or free QModMaster)	Read raw registers, count polling errors
Wireshark	Failover timing (1.1.5-2.5)

Item	Used for
SSH client (PuTTY) to the Juniper EX4100-H CLI	Junos version, port status, VLANs, spanning tree, config export
Windows Command Prompt	Ping tests
Loop calibrator (mA / V source and measure)	Analog inputs and outputs (only if analog IO is added; the E1210 is digital-input only)
DMM	Continuity, aux contact checks, DI verification
Tape measure, camera, stopwatch	SPD lead length, photo evidence, alarm and SCADA timing

For the PQMs and breakers, also bring Schneider ION Setup, ABB Ekip Connect, and a secondary injection test set or calibrated handheld power meter.

Documents

- Approved submittals: IO list, channel settings, network drawing and IP schedule, switch configuration backups, PQM settings (volts mode, PT/CT ratios, nominals), PM8000 and ABB Ekip Modbus maps
- ioLogik user manual (Modbus address table) for each module model
- SPD cut sheets and installation instructions
- SCADA points list
- Printed field checklists (`python run_tests.py checklists`, or the record tables in this procedure)

1.1.5-1 MOXA IO test

Every IO channel is checked by forcing a known condition at the field end and reading the result in ioAdmin, with Modbus Poll as a second, register-level reading.

This skid uses a MOXA ioLogik E1210: 16 digital inputs, no outputs or analog channels. Steps 1.1.5-1.2, 1.1.5-1.3 and 1.1.5-1.4 are recorded N/A unless other IO is added; they stay here for skids that have it.

1.1.5-1.5 Channel configuration

1. Connect the laptop to the skid switch. Run ioSearch and confirm every module appears at its scheduled IP.
2. Record each module's model and firmware version. Compare to the approved submittal.
3. Open each module in ioAdmin. For every channel, compare mode, range, filter and alarm settings to the IO list.
4. Export the module configuration and save it with the test records.
5. Pass: every module found, firmware matches, every channel setting matches.

1.1.5-1.1 Digital inputs

1. Open the module's I/O status view in ioAdmin.
2. At the field device or terminal, apply the ON (closed) condition. Confirm the channel shows 1 within 30 s.
3. Apply the OFF (open) condition. Confirm the channel shows 0.
4. Optional cross-check: in Modbus Poll, read function code 02 (discrete inputs) starting at address 0, quantity 16. On the E1210, addresses 0 to 15 are DI-00 to DI-15. Confirm the same values.
5. Pass: both states read correctly.

1.1.5-1.2 Digital outputs

1. Confirm with the lead that the output is safe to operate.
2. In ioAdmin, set the output ON. Confirm the field device responds and the output reads back 1.
3. Set the output OFF. Confirm the device de-energizes and reads back 0.
4. Leave every output OFF when done.
5. Pass: readback matches the command and the device responds both ways.

1.1.5-1.3 Analog inputs

1. Connect the calibrator in source mode to the channel.
2. Inject each test point: 0, 25, 50, 75 and 100% of the signal range (4, 8, 12, 16, 20 mA for 4-20 mA).

3. At each point, let the reading settle, then record 5 readings from ioAdmin (or Modbus Poll) a few seconds apart.
4. Calculate the expected value, the average error and the spread using the formulas below.
5. Pass: error within ±0.5% of span and spread within 0.25% of span at every point.

Expected engineering value at a test point:

```
E_{exp} = E_{min} + (E_{max} - E_{min}) \times \frac{pct}{100}
```

Engineering value from a raw register reading:

```
E = E_{min} + \frac{R - R_{min}}{R_{max} - R_{min}} \times (E_{max} -
E_{min})
```

Error and spread, both in % of span:

```
Error\% = \frac{E_{avg} - E_{exp}}{E_{max} - E_{min}} \times 100 \qquad
Spread\% = \frac{E_{high} - E_{low}}{E_{max} - E_{min}} \times 100
```

Worked example: BUS_TEMP is 4-20 mA = 0-150 °C. At 12 mA (50%) the expected value is 75.0 °C. Readings average 75.3 °C, so error = 0.3 / 150 × 100 = +0.20% (pass). Readings range 75.1 to 75.4 °C, so spread = 0.3 / 150 × 100 = 0.20% (pass).

1.1.5-1.4 Analog outputs

1. Confirm the output is safe to drive through its full range.
2. Connect the calibrator in measure mode to the output terminals.
3. In ioAdmin, command each test point (0, 25, 50, 75, 100%). Record the measured signal.
4. Error % = (measured − expected signal) / signal span × 100.
5. Pass: within ±0.5% of span at every point.

1.1.5-1.7 Alarm status logging

1. Open the alarm log on the SCADA or HMI (or keep ioAdmin's I/O status view open).
2. For each alarm point, trigger the alarm at the field end and start the stopwatch. Record the time the alarm appears.
3. Return the point to normal. Record the time the return-to-normal appears.
4. Screenshot the alarm log showing both entries with timestamps.

5. Pass: every alarm and return-to-normal is logged.

1.1.5-1.6 Communication polling over Ethernet

1. In Modbus Poll, open one read definition per module covering all its configured points. Set the scan rate to 1000 ms.
2. Reset the counters and run for 10 minutes (or the agreed duration).
3. Record Tx count and error count for each module. Screenshot the window.
4. Pass: 0 errors (or the limit agreed with the CxA).

IO record sheet

All 16 points are on IO-01 (ioLogik E1210), Modbus function 02, address = channel number. Contact type is the state in normal operation: NO (normally open) reads 0 when healthy, NC (normally closed) reads 1. Before the alarm test, record a healthy baseline with the skid in normal operation; every alarm point must read its normal value, or the wiring or contact type must be resolved first.

Ch	Tag	Description	Contact	Normal read	Baseline read	ON (1) read	OFF (0) read	Alarm logged	SCADA match	Pass/Fail	Initials
DI0	AUTO_MODE	Auto mode	NO	status				N/A			
DI1	MANUAL_MODE	Manual mode	NO	status				N/A			
DI2	MAIN_A_PREF_SOURCE	Main-A preferred source	NO	status				N/A			
DI3	MAIN_B_PREF_SOURCE	Main-B preferred source	NO	status				N/A			
DI4	125VDC_S2_FAIL	125VDC S2 fail	NO	0							
DI5	125VDC_S1_FAIL	125VDC S1 fail	NO	0							
DI6	240VAC_S_FAIL	240VAC S fail	NC	1							
DI7	SPD_ALARM	SPD	NC	1							
DI8	XFMR1_ALARM	Transformer 1 alarm	NO	0							
DI9	XFMR2_ALARM	Transformer 2 alarm	NO	0							
DI10	MAIN_A_LOCKOUT	Main A lockout	NC	1							
DI11	MAIN_B_LOCKOUT	Main B lockout	NC	1							

Ch	Tag	Description	Contact	Normal read	Baseline read	ON (1) read	OFF (0) read	Alarm logged	SCADA match	Pass/Fail	Initials
DI12	MAIN_A_ERMS_ON	Main A ERMS on	NC	1							
DI13	MAIN_B_ERMS_ON	Main B ERMS on	NC	1							
DI14	ATC_READY	ATC ready for operation	NO	status				N/A			
DI15	SPARE_DI15	Spare	-	-	N/A			N/A	N/A		

Paired points check: AUTO_MODE and MANUAL_MODE should never both read 1, and neither should MAIN_A_PREF_SOURCE and MAIN_B_PREF_SOURCE. Confirm this when operating the selector switches.

1.1.5-2 Network switch test

The skid switches are Juniper EX4100-H. Switch checks use the Junos CLI over SSH with a read-only login; timing checks use Command Prompt and Wireshark. Interfaces are named like `ge-0/0/0` (1G), `mge-0/0/0` (multigig) and `xe-0/2/0` (10G uplink).

1.1.5-2.1 Configuration and firmware

1. SSH to each switch. Run `show version` and record the model and Junos version. Compare to the submittal.
2. Run `show configuration | display set | no-more` and save the output as the running config.
3. Compare it to the approved config file (a text compare tool such as WinMerge makes differences obvious). Ignore comment lines and the `set version` line.
4. Run `show chassis alarms` and `show system alarms`; record any active alarms.
5. Pass: version matches, no config differences, no unexplained alarms.

1.1.5-2.3 Port status

1. Run `show interfaces terse` for admin and link state of every interface.
2. For each port on the network drawing, run `show interfaces ge-0/0/0 | match "Link|Speed"` (using that port's name) to record link state and speed.
3. Optional: `show poe interface` to confirm PoE status on any powered devices.
4. Pass: every port matches the drawing; unused ports are disabled or down as designed.

1.1.5-2.2 VLAN tagging

1. Run `show vlans` to list each VLAN name, ID and member interfaces.
2. Run `show ethernet-switching interface` to see each port's VLANs and whether each is tagged or untagged. For trunks, confirm the native VLAN with `show configuration interfaces xe-0/2/0` (using the port's name).
3. Compare mode (access or trunk), VLAN members and native VLAN to the network drawing.
4. Isolation check: from the laptop on one VLAN, ping a device on a different VLAN. It must fail unless the design routes between them. Then move to the other VLAN and confirm the device answers.
5. Pass: every port matches and isolation behaves as designed.

1.1.5-2.4 Ping response

1. For each device on the IP schedule (IO modules, PLC, HMI, meters, switches), run in Command Prompt:

```
ping -n 100 192.168.127.101
```

2. Record packets sent, received, lost, and the average and maximum round trip from the summary.
3. Count only lines containing `TTL=` as replies. Windows counts "Destination host unreachable" as received, so that line is a failure.
4. Pass: 0% loss and average round trip at or below 10 ms.

1.1.5-2.5 Redundant link failover time

A plain continuous ping only resolves about 1 second, which is too coarse to measure recovery accurately. Use a fast Modbus poll of the E1210 captured in Wireshark instead. The pass limit depends on the redundancy protocol in the network design (RSTP, MSTP or ERPS); agree it with the CxA before testing.

1. Confirm every redundant link is connected. On each switch, record the healthy state with `show spanning-tree interface` (RSTP/MSTP) or `show protection-group ethernet-ring interface` (ERPS).
2. In Modbus Poll, read one register from a module on the far side of the link under test. Set the scan rate to 20 ms.

3. Start a Wireshark capture on the laptop's Ethernet port with the display filter `mbtcp && ip.src == <module IP>`.
4. Let it run 10 s, then disconnect the link. Wait 15 s, then reconnect it. Wait another 15 s and stop the capture.
5. In Wireshark, add a column for `frame.time_delta_displayed` and sort it largest first. The largest gap near the disconnect time and the largest near the reconnect time are the two measurements.
6. Interruption = largest gap − normal scan interval. Save the capture file.
7. Repeat 3 times per link; record the worst result.
8. After the test, re-run the spanning-tree or ring command and confirm the ports returned to their healthy roles. Pass: interruption at or below the agreed limit on both break and restore.

1.1.5-2.6 SCADA network integration

1. For each point on the SCADA points list, make the change at the field end (toggle an input, or inject a value with the calibrator).
2. Read the field value in ioAdmin and the value on the SCADA screen. Time how long SCADA takes to update.
3. Pass: values match and the update is within the listed time (placeholder 2 s).

SCADA record sheet

Point names are from the DI list; replace them with the SCADA screen's tag names if they differ. Normal is what SCADA should show with the skid healthy.

Ch	SCADA point name	Contact	Normal	Field value	SCADA value	Update time (s) / max	Match	Initials
DI0	AUTO MODE	NO	status			____ / 2		
DI1	MANUAL MODE	NO	status			____ / 2		
DI2	MAIN-A PREF. SOURCE	NO	status			____ / 2		
DI3	MAIN-B PREF. SOURCE	NO	status			____ / 2		
DI4	125VDC S2 FAIL	NO	0			____ / 2		
DI5	125VDC S1 FAIL	NO	0			____ / 2		
DI6	240VAC S FAIL	NC	1			____ / 2		

Ch	SCADA point name	Contact	Normal	Field value	SCADA value	Update time (s) / max	Match	Initials
DI7	SPD	NC	1			____ / 2		
DI8	XFMR1 ALARM	NO	0			____ / 2		
DI9	XFMR2 ALARM	NO	0			____ / 2		
DI10	MAIN A LOCKOUT	NC	1			____ / 2		
DI11	MAIN B LOCKOUT	NC	1			____ / 2		
DI12	MAIN A ERMS ON	NC	1			____ / 2		
DI13	MAIN B ERMS ON	NC	1			____ / 2		
DI14	ATC READY FOR OPERATION	NO	status			____ / 2		

Network record sheet

Step	Device / port / link	Expected	Measured	Pass/Fail	Initials
2.1	SW-01 Junos version				
2.3	SW-01 ge-0/0/0 (IO-01)	up, 100 Mbps			
2.2	SW-01 ge-0/0/0	access, VLAN 10			
2.4	IO-01 192.168.127.101	0% loss, ⩽ 10 ms avg			
2.5	RL-01 break	⩽ agreed limit			
2.5	RL-01 restore	⩽ agreed limit			

1.1.5-2 PQMs and breakers (read and verify)

The skid network also carries three Schneider PowerLogic PM8000 meters (PQM1, PQM2, PQM3) and four ABB breakers with Ekip Com Modbus TCP (MAIN-A, AUTOXFMR, ACC-UPS, MAIN-B). These steps verify them without the test tool. They are read-only: PQM settings are programmed in ION Setup and breaker protection in Ekip Connect before testing, and nothing here changes a setting or operates a breaker remotely.

1.1.5-2.7 Device communication

1. Confirm each device answers `ping -n 100 <IP>` with 0% loss.

2. In Modbus Poll, connect to each device at its IP, port 502 and unit ID. Read one known register (for a PQM, Frequency: map register 3110, Modbus Poll address 3109, FLOAT32 big-endian; for a breaker, its status register) with a 1000 ms scan rate for 10 minutes.
3. Record Tx and error counts. Pass: 0 errors.

1.1.5-2.8 PQM configuration readback

1. Connect to each PQM in ION Setup. Open the Setup Assistant and record volts mode, PT primary and secondary, CT primary and secondary, nominal voltage, nominal frequency, IP address, Modbus unit ID and firmware version. Firmware must be 004.007.001 on all three PQMs.
2. Compare each value to the drawings and the settings list. PQM2 and PQM3 should match the template loaded from PQM1 except for any per-meter differences shown on the drawings.
3. Confirm CT and PT polarity are Normal on every phase unless the drawings call for an inversion.
4. Optional Modbus cross-check: in Modbus Poll, read the registers below with function 03 and compare them to ION Setup. Addresses are the map register number minus 1.
5. Save each meter's configuration from ION Setup into the run folder.
6. Pass: every setting matches.

Setting	Map register	Modbus Poll address	Type	Values
Meter name	30	29	Text, 20 registers	Set to PQM1 / PQM2 / PQM3
Firmware version	1717	1716	Text, 6 registers	004.007.001 on all PQMs (may display as V004.007.001)
Volts mode	54001	54000	UINT16	0 = 4W-WYE, 1 = DELTA, 2 = SINGLE, 3 = DEMO, 4 = 3W-WYE
CT polarity A / B / C	54002 to 54004	54001 to 54003	UINT16	0 = Normal, 1 = Inverted
Nominal frequency	54396	54395	UINT16	0 = 60 Hz, 1 = 50 Hz

Setting	Map register	Modbus Poll address	Type	Values
PT polarity V1 / V2 / V3	54587 to 54589	54586 to 54588	UINT16	0 = Normal, 1 = Inverted
PT primary / secondary	56001 / 56003	56000 / 56002	INT32	Volts
CT primary / secondary	56005 / 56007	56004 / 56006	INT32	Amps

Voltage, current, active power (in watts), power factor and frequency are FLOAT32 at map registers 3000 to 3150; nominal voltage and current are not in the Modbus map, so check those in ION Setup only.

1.1.5-2.9 PQM measurements vs reference

1. Apply a known source: secondary injection test set, or a stable load measured with a calibrated handheld power meter.
2. For each PQM, record voltage A-B, B-C, C-A, current A, B, C, total kW, power factor and frequency from the meter display (or ION Setup real-time screen), and the reference reading for each.
3. Error % = (meter − reference) / reference × 100.
4. Pass: each error within the agreed tolerance (placeholder 1% of reading; 2% for kW and PF; 0.1% for frequency).

1.1.5-2.10 Breaker status

1. For each breaker, read open/closed, tripped and racked position in Ekip Connect (or Modbus Poll using the ABB System Interface register map) and compare to the breaker's local indicators.
2. Operate the breaker by its normal local control (not remotely) and confirm the reported state follows. Return it to its required state.
3. Confirm SCADA shows the same state for each breaker.
4. Pass: reported state matches the equipment in every state checked.

PQM and breaker record sheet

Device	Check	Expected / reference	Reading	Error %	Pass/Fail	Initials
PQM1	Settings match (ION Setup)	per drawings		N/A		
PQM1	Voltage A-B					
PQM1	Current A					
PQM2	Settings match (ION Setup)	per drawings		N/A		
PQM2	Voltage A-B					
PQM3	Settings match (ION Setup)	per drawings		N/A		
PQM3	Voltage A-B					
MAIN-A	Status matches (open / closed)	local indicator		N/A		
AUTOXFMR	Status matches (open / closed)	local indicator		N/A		
ACC-UPS	Status matches (open / closed)	local indicator		N/A		
MAIN-B	Status matches (open / closed)	local indicator		N/A		

1.1.5-3 SPD test

SPD checks are visual and meter-based, so the manual method is the same one the tool prompts for; only the recording differs.

1.1.5-3.1 Placement and mounting

1. Confirm each SPD is at the location shown on the drawings.
2. Confirm mounting is secure with no damage, cracking or heat discoloration. Photograph each SPD.

1.1.5-3.2 Grounding and connection lead length

1. Measure each SPD's connection and grounding leads from terminal to termination point.
2. Record the length and compare to the manufacturer's maximum (placeholder 12 in).
3. Also note sharp bends or loops, which add impedance even when length passes.

1.1.5-3.3 Status LED and monitoring module

1. Confirm the local status LED shows healthy per the SPD cut sheet.
2. Confirm the monitoring module is installed and fully seated.

1.1.5-3.4 Dry auxiliary alarm contact

1. With ioAdmin open (or a DMM across the contact if it isn't wired to a MOXA input), confirm the SPD contact (DI7, NC) reads 1 with the SPD healthy.
2. Simulate a fault per the manufacturer's method, typically removing the replaceable module. Confirm DI7 changes to 0 and the SPD alarm appears on SCADA.
3. Restore the SPD. Confirm DI7 returns to 1 and the alarm clears.

1.1.5-3.5 Continuity of surge suppression circuit

1. De-energize and apply LOTO. Verify absence of voltage.
2. Measure continuity from the SPD's line terminals to the protected conductors and from the ground terminal to the ground bus.
3. Pass: each reading at or below the spec limit (placeholder 1.0 Ω).

SPD record sheet

SPD	Placement / mounting	Lead length (in) / max	LED healthy	Monitor module	Aux: normal / fault / normal	Continuity (Ω)	Pass/Fail	Initials
SPD-01		____ / 12						

Cross-checking the tool, recording and sign-off

On the first skid, repeat a sample of automated results by hand; if every sample agrees, the tool is verified for that skid design.

Suggested verification sample

Tool test	Manual check	Agreement required
1.1.5-1.1 DI	2 inputs per module, both states	Same state

Tool test	Manual check	Agreement required
1.1.5-1.3 AI	N/A on the E1210 (digital only). If analog IO is added: every AI at 50%, one AI at all five points	Engineering value within 0.1% of span
1.1.5-1.6 Polling	10 min Modbus Poll run on one module	Both show 0 errors
1.1.5-2.3 / 2.2	Every port's status and VLANs from the Junos CLI	Identical
1.1.5-2.4 Ping	3 devices by Command Prompt	Both 0% loss; averages within a few ms
1.1.5-2.5 Failover	One link by Wireshark	Within 5 ms of the tool
1.1.5-3.4 SPD aux	One SPD	Same state sequence

If a manual and automated result disagree, stop, find the cause (config address, scaling, ifIndex mapping, meter error) and re-run both before continuing.

Recording

- Fill the record sheets in ink; mark each line Pass, Fail or N/A, with initials and date.
- Save exports, screenshots, Wireshark captures and photos in the skid's run folder so they appear in the report's Attachments list.
- Any failure goes on the Open_Items sheet with corrective action and retest date.

Sign-off

The tester (manufacturer), the Owner's witness and the CxA sign the report cover. When manual results replace or verify automated ones, state that in the report and attach the signed record sheets.